21-242 Recitation 12

Span, Linear Independence, Bases — Q4. Can appear in a linearly independent list?

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Can $\vec{0}$ appear in a linearly independent list?

Course-level solution

No. If $\vec{v}_j = \vec{0}$, then $1 \cdot \vec{v}_j = \vec{0}$ is a nontrivial linear combination of the list (or, in sequential language, $\vec{0}$ always lies in the span of the preceding vectors, including the empty span when $j = 1$).
So any list containing $\vec{0}$ is linearly dependent.